



Higher Technical School of Telecommunications Engineering and Systems



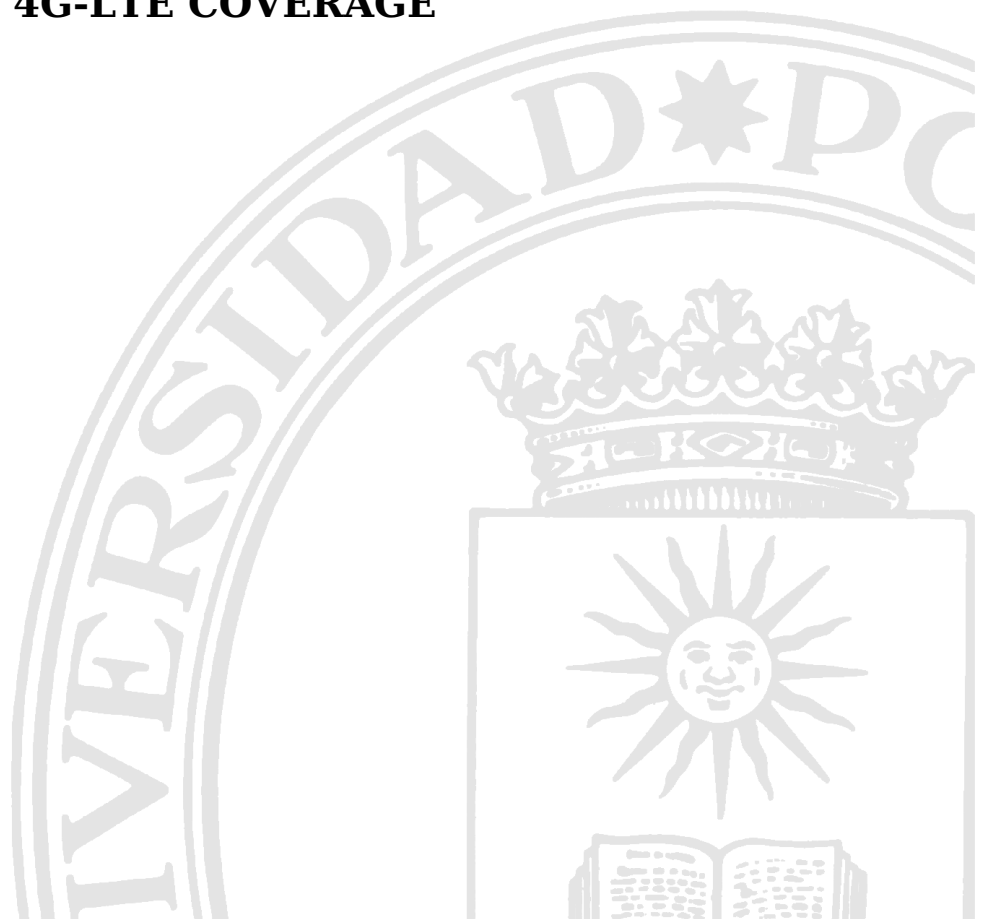
RADIOCOMMUNICATION SYSTEMS

PRACTICE 4

4G-LTE COVERAGE

Surname and Name:

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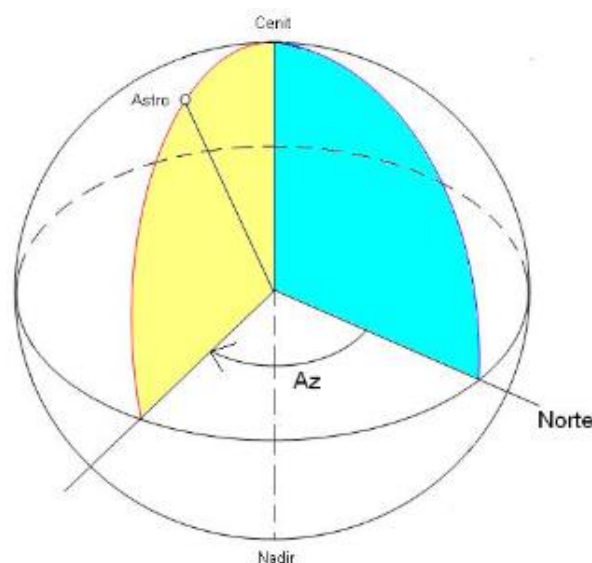


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2. Design and Configuration Parameters

The network uses the LTE standard in the 1800 MHz band with a 20 MHz channel bandwidth. Strict parameters were defined for both the downlink and the user terminals. Each base station transmits 32 W through 65-degree sector antennas with 17.5 dBi gain. The simulated mobile terminal is 1.5 m high and has a critical reception threshold of -112 dBm, which defines the service-coverage limit.

Theoretical note

An LTE system divides the radio channel into multiple subcarriers and dynamically allocates time-frequency resources according to user demand and signal quality. Multiple-antenna techniques and adaptive modulation are used to maximize data rate while maintaining connectivity at the cell edge, which is a critical objective of this Xirio plan.

ESTACIÓN BASE	
Sectores	3
Banda de frecuencia	LTE 1800-20MHz
Potencia transmisor	32W
Perdidas del alimentador de antena(feeder)	0 dB
Parametros LTE	Por defecto Xirio
Antena	3G/4G/5G 17,5 dBi 65°
Altura antena máxima	30m
Inclinación mecánica	0-6°

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Terminal MOVIL	
Antena	2dBi Omni
Altura antena	1.5m
Potencia	23dBm
Umbral Recepción	-112dBm

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Elementos principales de la arquitectura LTE

Red de acceso radio: E-UTRAN (Evolved UMTS Terrestrial Radio Access Network)

Consta de estaciones base, denominadas eNodeB o eNB

Funciones:

- Modulación y demodulación, codificación de canal
- Control de enlace radio: detección y corrección de errores
- Control de recursos radio: asignación, cambio, liberación
- Gestión de movilidad: procesamiento de medidas y traspasos
- Intercambio de tráfico entre móvil y núcleo de red
- Otras: cifrado (datos y señalización)

Núcleo de red: EPC (Evolved Packet Core)

eNodeB	Evolved NodeB
MME	Mobility Management Entity
SGW	Serving Gateway
PGW	PDN Gateway
HSS	Home Subscriber Server
PCRF	Policy & Charging Rules Function
UE	User Equipment

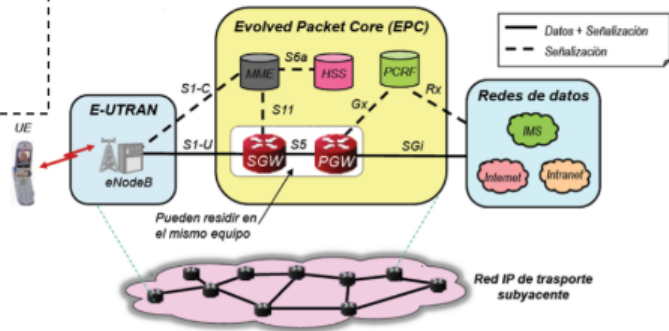


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3. Site Definition

Three strategic locations were selected to provide contiguous coverage and form a logical topology. The following pages describe the geographical coordinates, tower heights and sector-specific angular configurations for each base station.

Before implementing the system in Xirio, an initial theoretical approximation was prepared using the terrain characteristics and the suburban and rural areas to be served.

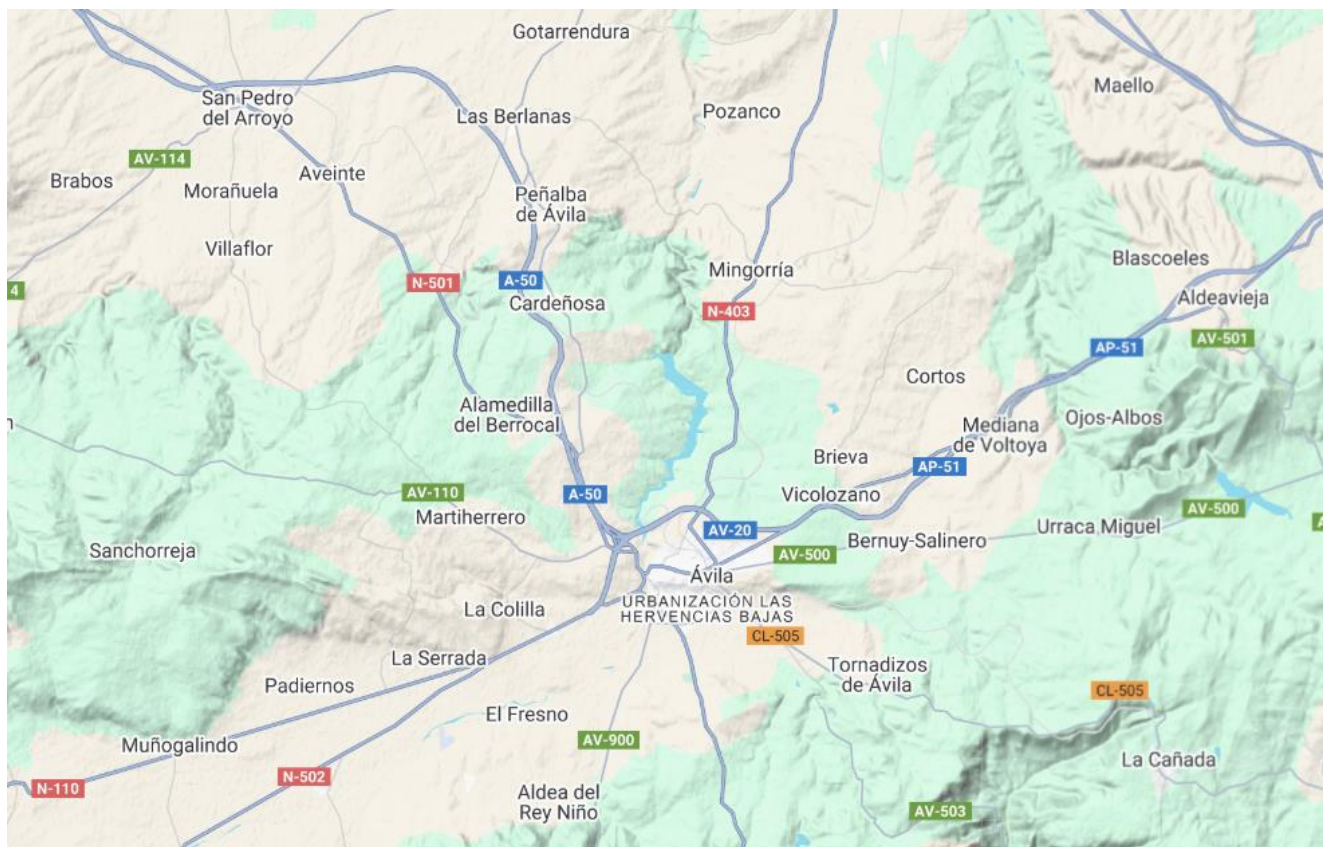


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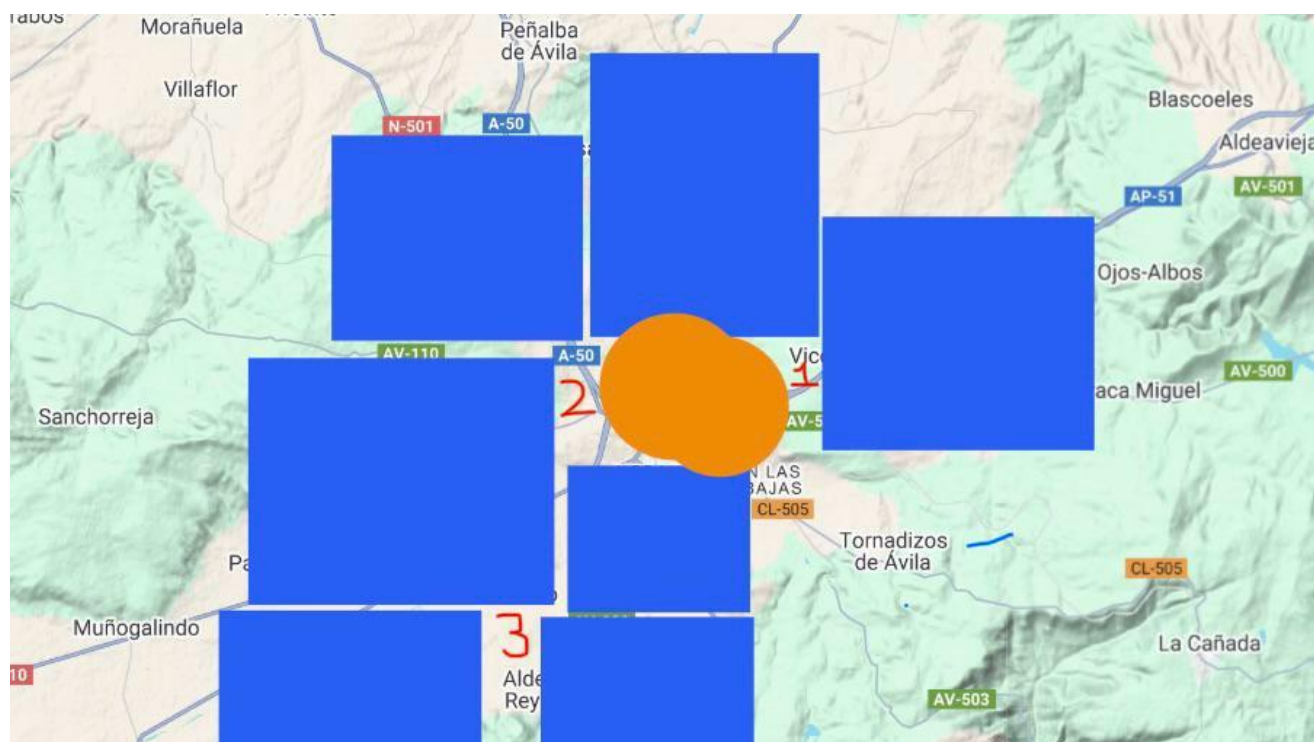


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3.1. Base Station 1

Site elevation: 1200 m

Sector 1 - Suburban

- Antenna azimuth: 240 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

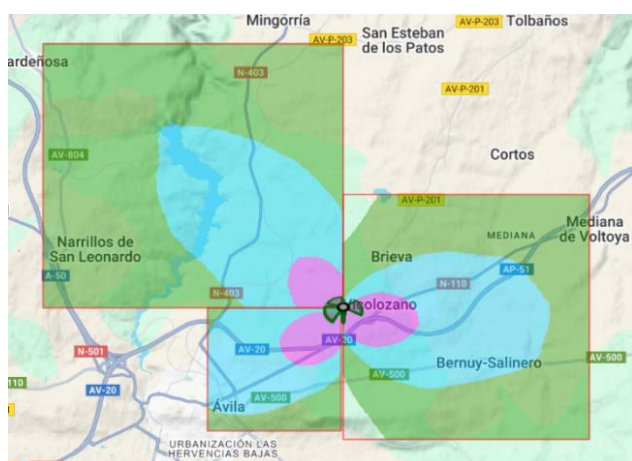


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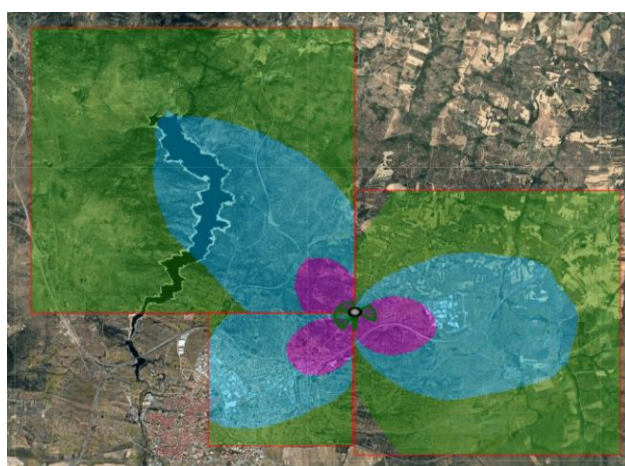


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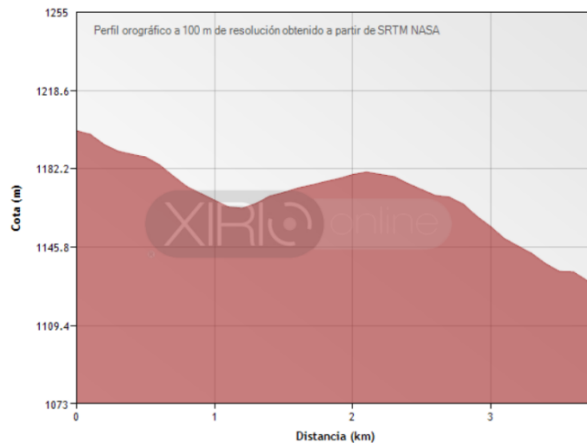


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Sector 2 - Rural

- Antenna azimuth: 325 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

Sector 3 - Rural

- Antenna azimuth: 90 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

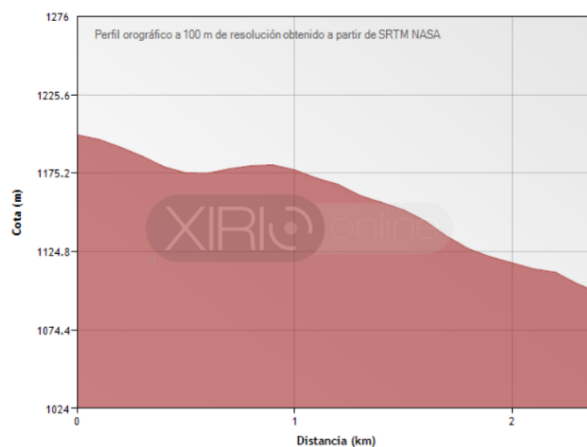


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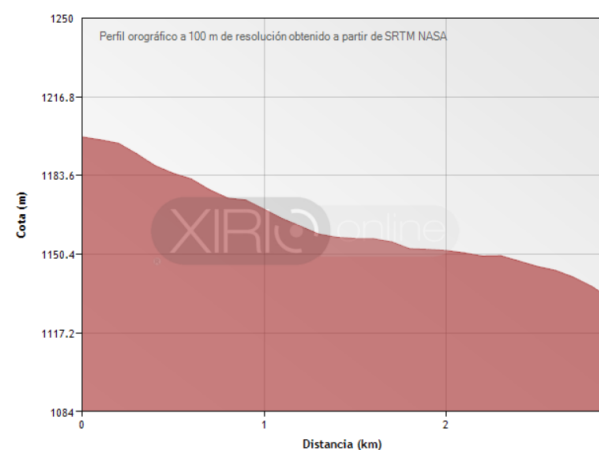


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3.2. Base Station 2

Site elevation: 1122 m

Sector 1 - Suburban

- Antenna azimuth: 110 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

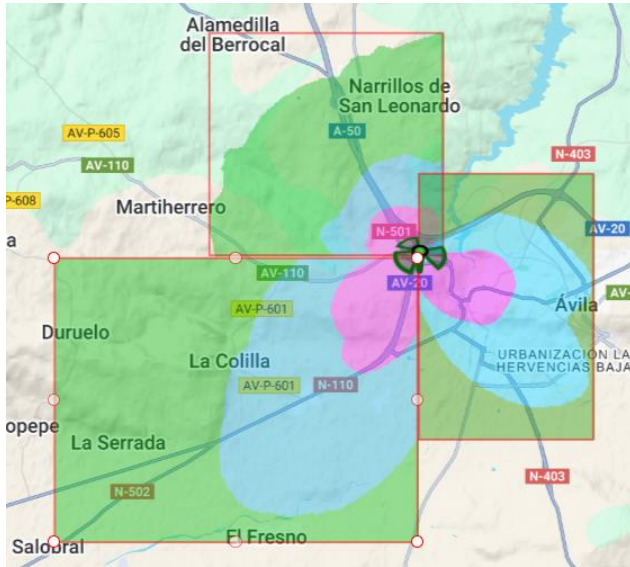


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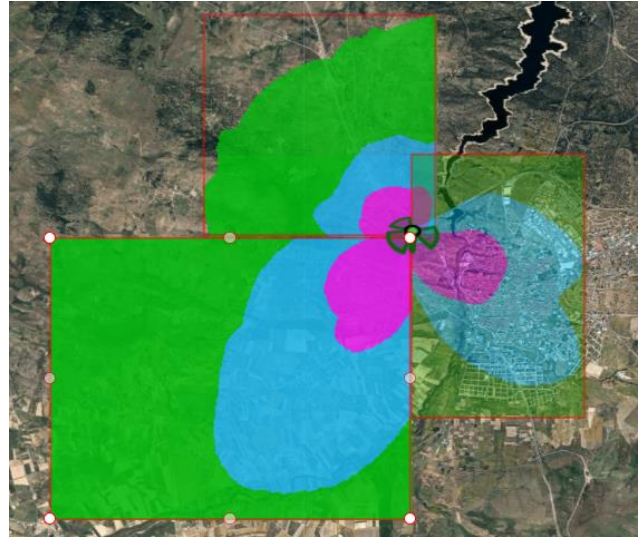


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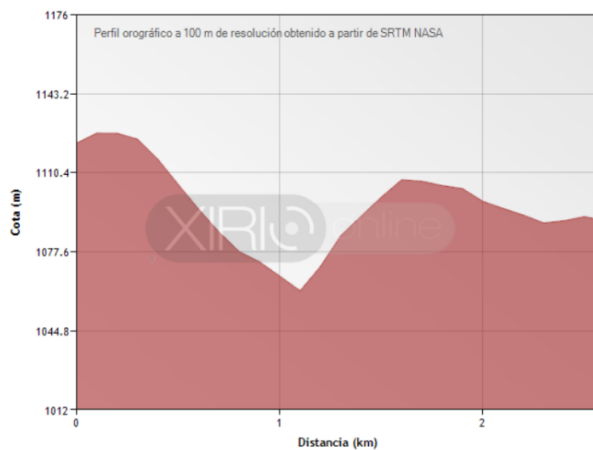


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Sector 2 - Rural

- Antenna azimuth: 320 deg
- Antenna height: 30 m
- Mechanical tilt: 10 deg
- Terrain profile and coverage statistics are shown in the original figures.

Sector 3 - Rural

- Antenna azimuth: 230 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

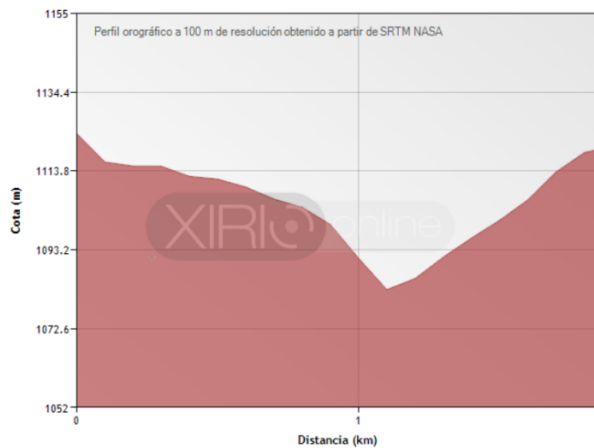


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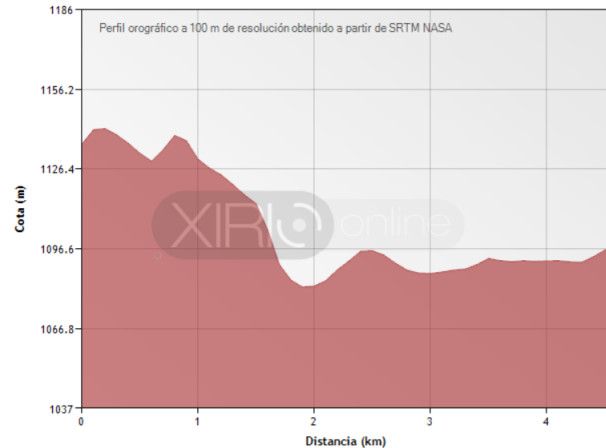


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3.3. Base Station 3

Site elevation: 1105 m

Sector 1 - Rural

- Antenna azimuth: 45 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

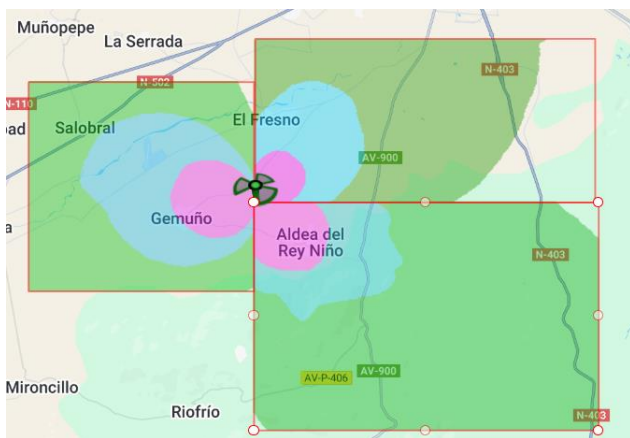


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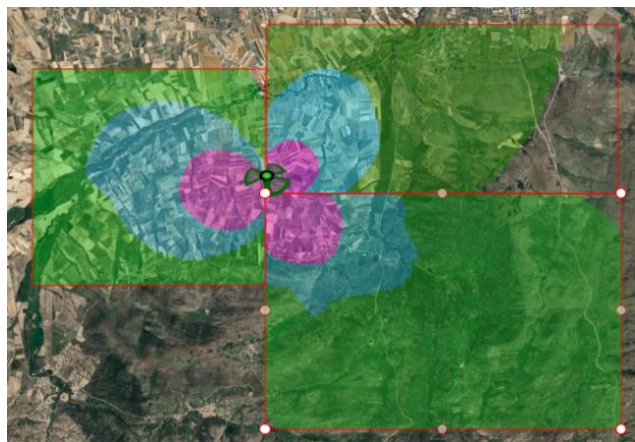


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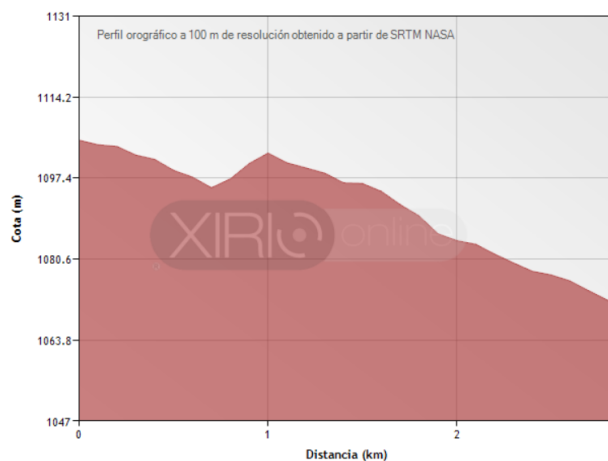


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Sector 2 - Rural

- Antenna azimuth: 270 deg
- Antenna height: 30 m
- Mechanical tilt: 3 deg
- Terrain profile and coverage statistics are shown in the original figures.

Sector 3 - Rural

- Antenna azimuth: 135 deg
- Antenna height: 30 m
- Mechanical tilt: 0 deg
- Terrain profile and coverage statistics are shown in the original figures.

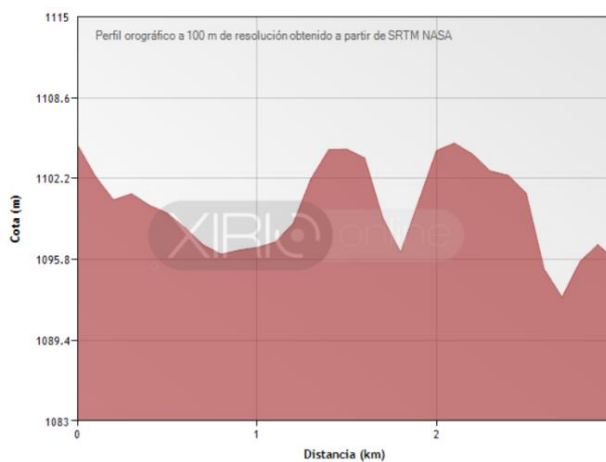


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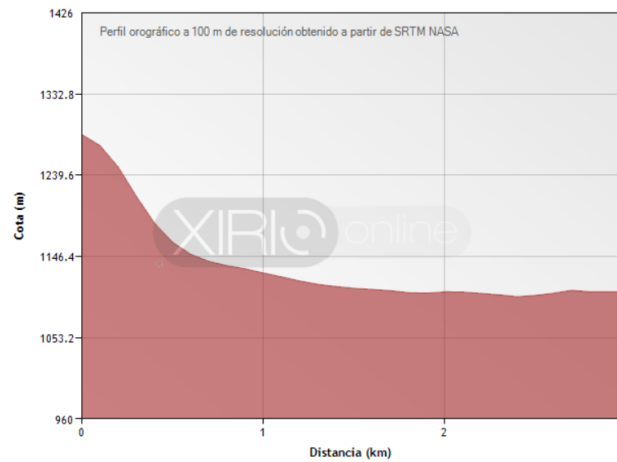


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3.4. Design Justification

The proposed design is based on four principles:

- Base-station placement: the sites form a perimeter around the urban center. This allows the city to be covered from the outside inward while also serving main roads and nearby rural areas.
- Adaptation to terrain profiles: Avila has high and highly irregular terrain. Most sectors use zero mechanical tilt to maximize range over relatively open areas. Two sectors use 10-degree and 3-degree tilt to compensate for terrain irregularities and provide service in nearby depressions and valleys that would otherwise be shadowed.
- Total covered area: the selected configuration maximizes the useful service area. Coverage statistics show that almost the entire suburban/rural calculation area is served, with few gaps between cells and sector ranges of roughly 20 to 25 km from different directions.
- Coverage quality: received signal exceeds the -112 dBm sensitivity threshold throughout the target area. Antenna orientation also creates sufficient overlap for handover between stations without causing critical interference.

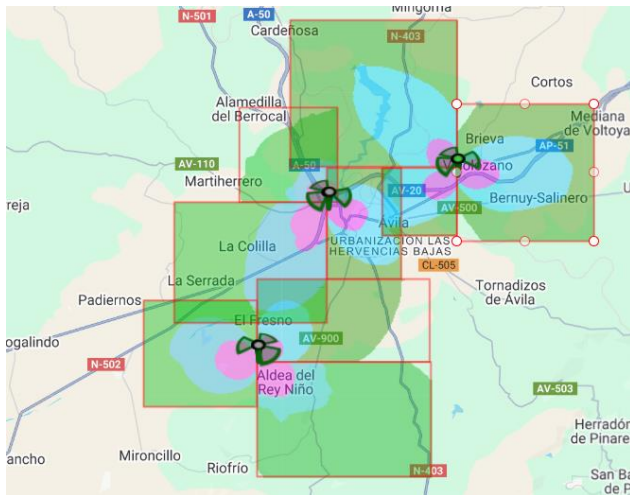


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4. Multi-Transmitter Study Results

A combined multi-transmitter calculation was carried out using the three base stations and their nine sectors in a single simulation. The objective was to verify that the -112 dBm minimum reception threshold is met by the mobile terminal throughout the study area around Avila.

The combined coverage plot shows that the site layout illuminates the target area effectively, including the urban center and the main access roads, without significant dark zones between base stations.

■	Ávila1_S1				
■	Ávila1_S2				
■	Ávila1_S3				
■	Ávila2_S1				
■	Ávila2_S2				
■	Ávila2_S3				
■	Ávila3_S1				
■	Ávila3_S2				
■	Ávila3_S3				

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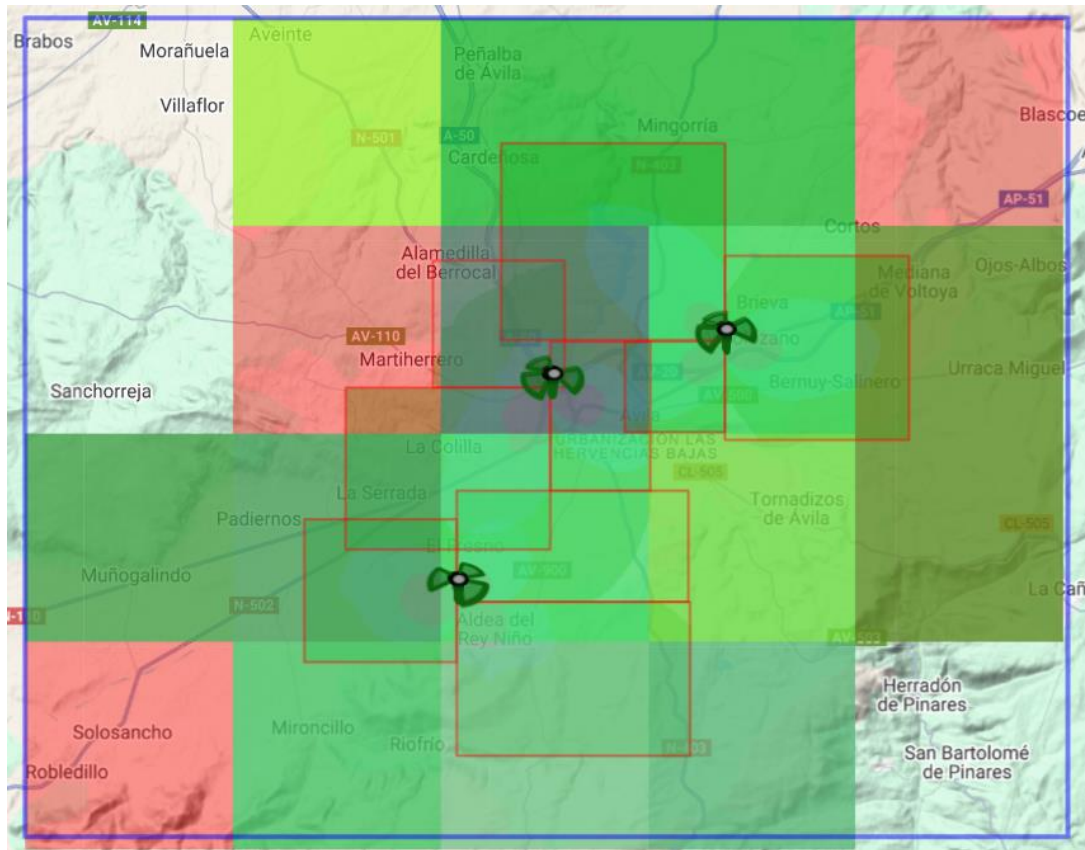


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The global simulation statistics show solid network coverage. At the critical design threshold of -112 dBm, located between the -110 and -115 dBm result columns, approximately 70% of the total calculation area is covered: 804 surface units out of 1,149.

The remaining 30% mainly corresponds to the corners of the calculation rectangle, which contain complex terrain or remote rural areas outside the primary service objective. In the area of interest - the urban center and national roads - robust levels are achieved. Approximately 25% of the territory exceeds -95 dBm, supporting high-speed services.

5. Network Quality Analysis

5.1. Best-Server Analysis

The results identify dominant sectors such as sector 3 of base station 1, with 229 surface units, and sector 2 of base station 3, with 172 units. These sectors provide most of the open rural coverage. Urban sectors cover smaller, more controlled areas, which is appropriate for capacity management.

- ☐  Señal RSRP
- ☒  **Mejor servidor**
- ☐  Solapamiento

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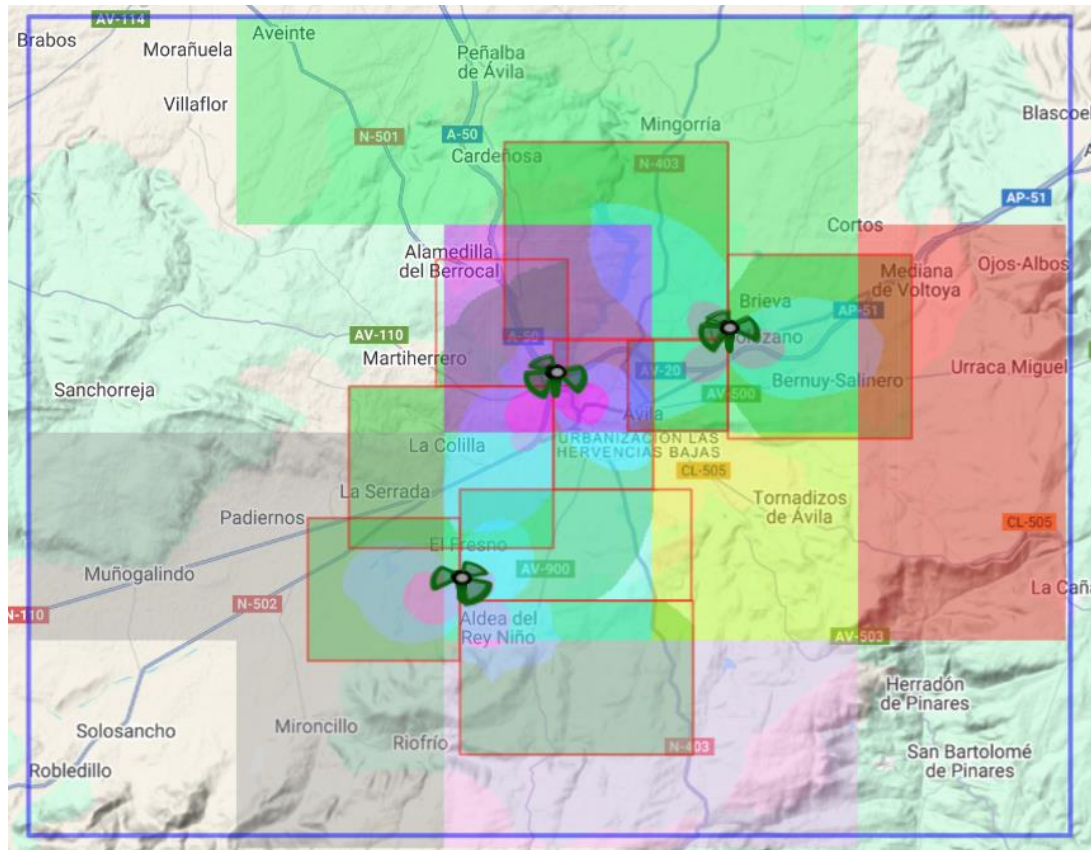


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Mejor Servidor	Código	Topónimo	S Total	S>=-120dBm	S>=-115dBm	S>=-110dBm	S>=-105dBm	S>=-100dBm	S>=-95dBm	S>=-90dBm	%S>=-120dBm	%S>=-115dBm	%S>=-110dBm	%S>=-105dBm	%S>=-100dBm	%S>=-95dBm	%S>=-90dBm
SG1_S3			1149,73549	229,947099	229,947099	229,947099	172,460324	57,4867747	57,4867747	57,4867747	20	20	20	15	5	5	5
SG2_S1			1149,73549	57,4867747	57,4867747	57,4867747	57,4867747	0	0	0	5	5	5	5	0	0	0
SG1_S3			1149,73549	114,973549	114,973549	114,973549	114,973549	0	0	0	10	10	10	10	10	0	0
SG3_S2			1149,73549	172,460324	172,460324	172,460324	172,460324	57,4867747	57,4867747	0	15	15	15	15	5	5	0
SG2_S3			1149,73549	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	5	5	5	5	5	5	5
SG1_S2			1149,73549	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	57,4867747	5	5	5	5	5	5	5
SG3_S3			1149,73549	114,973549	114,973549	114,973549	114,973549	114,973549	57,4867747	57,4867747	10	10	10	10	10	5	5

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5.2. Overlap-Area Analysis

Effective overlap is visible along routes such as the A-50 motorway and the N-110 road. In these transition zones, a terminal receives a valid signal from at least two sectors at the same time.

Distance measurements on the map confirm that the transition strips are wider than 300 m, meeting the design criterion.



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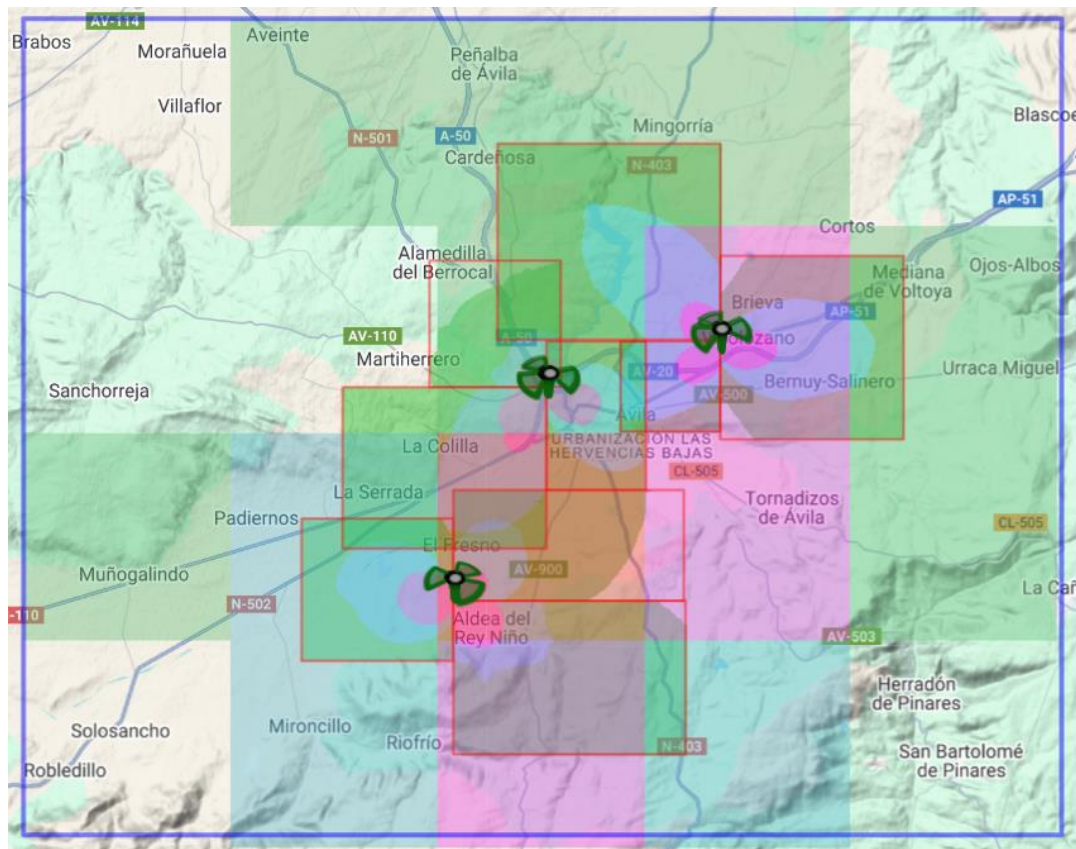


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6. Conclusions

The design and simulation of the 4G network for Avila met the main objectives. Three base stations provide service to the urban center, surrounding areas and principal roads, while maintaining sufficient received power above the -112 dBm threshold across most of the target zone. The design could be refined further to provide even stronger guarantees, but it successfully resolves the main challenges of the practice.

Planning was labor-intensive because of Avila's high and irregular terrain. Standard antenna settings were not sufficient; repeated trial-and-error adjustments, especially to mechanical tilt, were required to prevent terrain from blocking the signal in critical areas.

The multi-transmitter study was initially abstract, and interpreting the resulting maps was one of the most difficult parts. The best-server and overlap plots ultimately made it possible to understand how the three sites cooperate to fill coverage gaps and allow users to move without losing connectivity.

Despite the terrain constraints and the complexity of the simulation tool, the final design is technically feasible and functional for the selected area. The practice provided practical knowledge of 4G-LTE coverage and helped connect theoretical concepts, such as the Okumura-Hata propagation model, with a real geographical planning problem.